

RESEARCH ESSAY + PREREGISTERED STUDY

The Ecological Compiler

Marine Nutrition, Maritime Connectivity, and the Persistence of Collective Capability

1,145

adjusted societies

1,100

registered resamples

4 / 4

gates passed / failed

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Registered verdict: **REJECTED** at the descriptive claim tier

Abstract

This paper began with a provocative causal story: migration into low-ultraviolet environments increased pressure for dietary vitamin D; fish supplied vitamin D and long-chain omega-3 fatty acids; those fatty acids increased dopamine and cognitive capacity; and those biological advantages helped produce European maritime and political development. Research does not support that chain. Human pigmentation reflects repeated, contingent interactions among ultraviolet radiation, genes, migration, diet, clothing, and shelter. EPA and DHA are not dopamine precursors. Randomized human evidence does not show a large general cognitive gain from omega-3 supplementation. Most decisively, a preregistered analysis of 1,145 D-PLACE societies does not find a robust, transportable relationship between fishing dependence and jurisdictional hierarchy. The adjusted ordered-logit coefficient is positive, but language-family and spatial block intervals include zero, and the coefficient reverses after the three largest language families are removed. The registered headline is rejected. Something better survives. Marine ecology changes several constraints at once. It can relieve nutritional deficiencies, concentrate defensible resources, support sedentism and storage, reduce transport costs, expand networks, and generate new organizational problems. These effects do not accumulate automatically. They persist when societies possess carriers that preserve, select, recombine, and transmit capability across generations. I call the coupled process an **ecological compiler**. The sea supplies affordances. Human bodies, technologies, organizations, and institutions determine what those affordances become. This framework explains why the same marine exposure can sustain an egalitarian fishing community, a hereditary hierarchy, a trading city, an empire, or no durable political expansion at all. The conclusion is therefore firm but bounded: fish did not make Europe cognitively superior, and fishing dependence is not a general cause of state complexity. Marine environments matter when their concrete resource and network properties are converted by historically specific institutions.

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I. The hypothesis that needed to fail

The original intuition had the appeal of a straight line:

*low UV -> lighter pigmentation -> vitamin-D pressure -> fish -> EPA/DHA -> dopamine -> planning
-> maritime power -> civilization.*

Almost every term points toward a real field of research. The arrows are the problem.

A chain can be locally plausible and globally false. Low ultraviolet exposure can change the adaptive value of cutaneous vitamin-D production. Fatty fish can supply vitamin D. DHA matters to neuronal membranes. Dopaminergic systems participate in motivation, learning, and action selection. Ships can reorganize trade, warfare, and state finance. None of those facts establishes the next arrow, much less the whole chain.

The companion manuscripts for *A New Political Science* gave me the right way to interrogate this problem. One manuscript treats politics as a structured constraint problem in which geography defines a prior over possible arrangements rather than a completed explanation of them. The other asks what carries a capability through time. A population, a language, an archive, an organization, and an institution can each preserve different parts of a past state. Historical persistence cannot be inferred from a correlation between then and now. The carrier has to be named and separated from its alternatives (Brown 2026a, 2026b).

That observation changes the question. The useful question is not whether fish caused European civilization. It is whether marine environments relaxed particular biological constraints, altered the topology of interaction, and created resources or routines that some social systems could retain. The resulting theory has to distinguish nutrition from navigation, individual capacity from collective memory, and ecological opportunity from institutional conversion.

It also has to survive comparison. A biochemical mechanism cannot recognize a continental mythology. If marine omega-3 exposure causes inherited collective superiority, then the prediction should travel across Arctic, Pacific, Atlantic, African, Asian, and American fishing populations. If it does not travel, the explanation is not biochemical in the proposed sense.

I therefore treated the original story as a claim to defeat. Before the primary analysis, I registered variables, estimands, controls, negative controls, resampling rules, transport tests, and eight noncompensatory gates. The data were allowed to return one of three answers: a robust association, a bounded pattern, or a rejection. They returned the third for the headline claim and the second for a narrower theory.

II. What the biology permits, and what it forbids

The strongest biological claim concerns constraint, not enhancement.

Human skin pigmentation is an adaptation shaped substantially by ultraviolet radiation. High-UV environments favored dark, eumelanin-rich protection, while low-UVB environments increased pressure to admit enough UVB for cutaneous vitamin-D production. This was not a single European switch. Depigmentation evolved more than once through different genetic pathways, and later syntheses place genes inside a contingent system that also includes migration, diet, clothing, shelter, and activity (Jablonski and Chaplin 2010; Jablonski 2021).

Put simply, latitude changed the problem. It did not dictate one solution.

Marine foods can be one solution because fatty fish and marine mammals can supply dietary vitamin D where cutaneous production is seasonally constrained. But populations can also change pigmentation, seek different foods, alter exposure through clothing or shelter, migrate, store food, or eventually use fortification. These responses are partially substitutable. A darkly pigmented population eating a vitamin-D-rich marine diet and a depigmented agricultural population obtaining more cutaneous vitamin D may solve part of the same physiological task through different concrete arrangements.

The Greenlandic Inuit case makes the point sharper. Fumagalli and colleagues found strong selection signals around fatty-acid desaturase genes in a population with a long history of PUFA-rich marine diets. The result is evidence for diet and

genome coevolution, not for a fixed cognitive benefit delivered by fish. A food environment does not act on an unchanged machine. Over time, the machine and the diet can change together (Fumagalli et al. 2015).

The dopamine claim requires a harder correction. EPA and DHA are not precursors of dopamine. Dopamine biosynthesis proceeds from tyrosine to L-DOPA through tyrosine hydroxylase, then from L-DOPA to dopamine. The pathway is regulated through enzyme activity, phosphorylation, feedback, cellular state, and vesicular handling (Daubner, Le, and Wang 2011). Omega-3 fatty acids can affect membrane composition and may modulate dopaminergic systems, especially under deficiency in animal models, but modulation is not precursor chemistry (Healy-Stoffel and Levant 2018).

The cognitive evidence is narrower still. A 2024 systematic review and dose-response meta-analysis covered 24 randomized studies and 9,660 middle-aged or older adults without dementia. In random-effects models, omega-3 intake was not associated with a detectable change in global cognition or five other domains. The authors found limited, nonlinear patterns for executive function under some durations and doses, with considerable heterogeneity (Suh et al. 2024). That is compatible with selective benefit or deficiency relief. It is not evidence for a general cognitive multiplier.

The distinction is basic:

deficiency relief != unbounded enhancement.

When a nutrient is below a threshold needed for ordinary development, obtaining it can expand the feasible set of learning, health, and action. Around adequacy, more need not produce a proportional cognitive gain. Nutrition helps define what an organism can sustain. It does not select a constitution, invent a ledger, build a port, or preserve a theorem.

This is the first proof gate. The biological literature permits a theory of constraint relaxation and context-sensitive neural modulation. It forbids the proposed fish-to-dopamine shortcut.

III. Study I: forcing the theory into data

The first empirical test asks the cheapest global question the theory can answer: does dependence on fishing predict political hierarchy across small-scale and preindustrial societies?

The data come from the current D-PLACE release of the Ethnographic Atlas. D-PLACE links cultural observations to society locations, environmental measures, language classifications, and source metadata. Its design also makes the central methodological danger explicit: societies near one another in space or on a language tree are not independent draws (Kirby et al. 2016).

The exposure is EA003, dependence on fishing, shellfishing, and large aquatic animals. It is coded in ten ordered categories from 0 to 5 percent through 86 to 100 percent of subsistence. The outcome is EA033, jurisdictional hierarchy beyond the local community. It runs from no authority above the community through four jurisdictional levels associated with large states. This outcome is not a measure of intelligence, moral worth, institutional quality, or “civilization.” It is a narrow ordinal measure of political scale.

The preregistered analysis used three proportional-odds models:

Model	Question	Registered adjustment
M0	What is the unadjusted association ?	Fishing only.

Model	Question	Registered adjustment
M1	Does fishing retain an association after measured ecological and subsistence alternatives?	Hunting, husbandry, agriculture, absolute latitude, logged distance to coast, annual temperature, temperature variance, precipitation predictability, net primary production, focal year, and six D-PLACE macroregion fixed effects.
M2	Does the fishing coefficient attenuate when plausible social compilers enter?	M1 plus settlement pattern, local community size, and population.

Gathering is the omitted share-like subsistence category in the fishing model. M2 variables can be mediators, confounders, or consequences, so M2 is not a clean mediation estimate. It is a registered pattern check.

The primary M1 sample contains 1,145 societies, 164 language-family clusters, 83 spatial blocks, six macroregions, and all five hierarchy levels. Missing language-family labels form society-specific singleton clusters rather than one invented ancestry bloc. The analysis uses exact source commits and hashes, clean pinned data worktrees, an analytic-gradient ordered-logit implementation, 300 language-family block bootstraps, 300 spatial block bootstraps, and 500 within-macroregion permutations. Every registered resample converged in the final run. Four binary threshold models test whether the proportional-odds coefficient hides a sign reversal.

The gates were intentionally strict. A positive point estimate was insufficient. The adjusted claim required a positive M1 coefficient, a language-family block interval excluding zero, and a one-sided within-macroregion permutation value no greater than 0.05. Separate gates required subsistence specificity, attenuation after social variables, transport outside Europe and across family exclusions, and sign stability across outcome thresholds. A failed necessary gate could not be averaged away by a good result elsewhere.

This design still has a ceiling. A language-family block bootstrap is not a phylogenetic generalized linear model. Recent simulations and reanalyses show that family fixed effects and clustered uncertainty can leave substantial ancestral dependence unresolved in Ethnographic Atlas research (Gershman and Mumladze 2025). The present study is therefore a stringent descriptive screen, not a causal or phylogenetic endpoint.

IV. The result: a signal that does not generalize

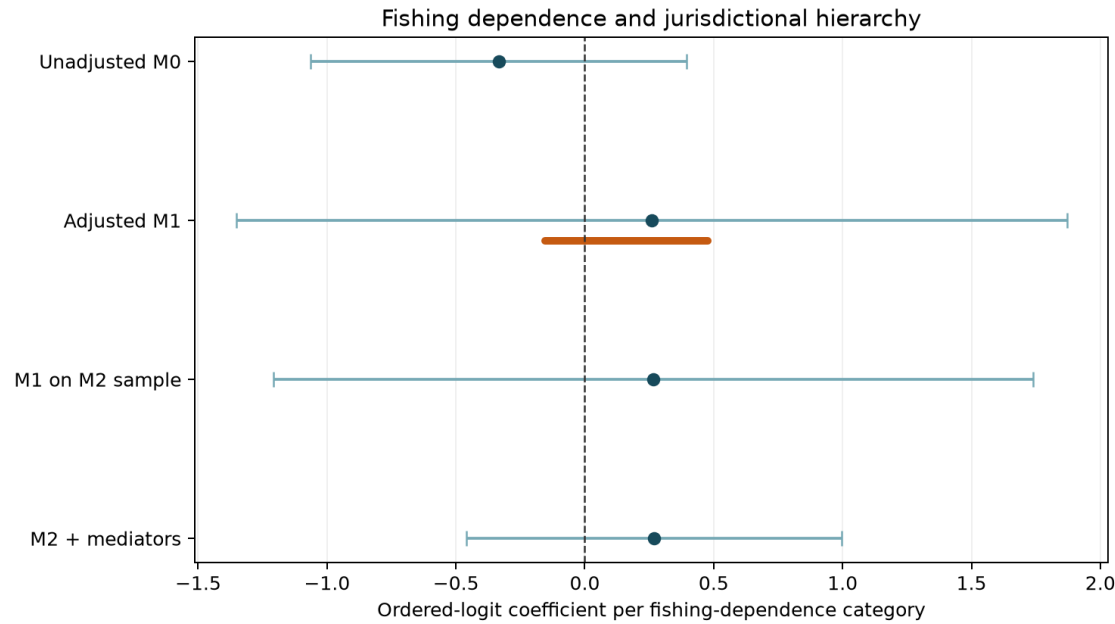


Figure 1. Registered ordered-logit estimates. The orange interval is the 300-draw language-family block interval for M1; thin intervals are optimizer-Hessian approximations and are not the primary uncertainty estimate.

The registered headline failed.

The unadjusted association runs in the opposite direction from the original theory. In M0, each higher fishing-dependence category is associated with a coefficient of -0.334 on jurisdictional hierarchy. After ecological, geographic, temporal, macroregional, and alternative-subsistence controls enter, the M1 coefficient becomes +0.259. That reversal is already a warning. The raw relationship is dominated by where fishing societies are, what other subsistence systems they use, and which broad historical regions they occupy.

The adjusted coefficient corresponds to an ordered-odds ratio of about 1.30 per fishing category, but the uncertainty needed for cross-cultural generalization crosses the null. The 300-draw language-family block interval is [-0.155, 0.478]. The 300-draw spatial-block interval is [-0.061, 0.436]. Both include zero by a meaningful margin.

The within-macroregion permutation test points the other way. Its one-sided value is 0.0020, with all 500 draws converged. Within the model’s broad geographic exchangeability assumption, the observed coefficient is unusual when fishing codes are shuffled. But that result cannot compensate for the ancestry-sensitive interval. The two tests ask different questions. The permutation result says the adjusted within-macroregion pattern is not easily reproduced by random reassignment of fishing dependence. The family bootstrap says the sign and magnitude do not generalize reliably across the language-family composition of the sample.

That tension is the study’s central empirical result. There is a structured association in these data. It is not a stable global law.

Table 1. Primary registered estimates

Model	Societies	Fishing coefficient	Standardized coefficient	AIC
M0, unadjusted	1,155	-0.334	-0.584	2932.32
M1, adjusted	1,145	+0.259	+0.447	2468.82
M1 on the M2 complete-case sample	467	+0.267	+0.480	962.76
M2, adding settlement and population variables	467	+0.267	+0.482	905.29

Fishing is more specific than two substitute exposures in the registered comparison. Its standardized M1 coefficient is +0.447, compared with +0.214 for hunting and -0.413 for gathering. Because the subsistence variables are compositional and share model terms, this is a limited contrast rather than independent proof of a marine mechanism. All four binary hierarchy thresholds also retain a positive fishing coefficient. These checks passed the specificity and ordinal-stability gates, although the latter uses optimizer-Hessian standard errors only as a sensitivity heuristic.

The proposed compiler pattern failed. On the same 467-society sample, adding settlement, community size, and population improves AIC by 57.47 points but leaves the fishing coefficient effectively unchanged: +0.267 before and +0.267 after. The social variables explain hierarchy-relevant variation, but this cross-section does not show that they carry the fishing association. Better fit is not attenuation, and it cannot substitute for the missing pathway evidence.

Transport is where the broad claim also breaks. The coefficient remains positive outside the European-labeled regions at +0.309. It remains positive in all six leave-one-macroregion-out refits, ranging from +0.128 to +0.352. But removing the three largest language families, Atlantic-Congo, Austronesian, and Afro-Asiatic, changes the coefficient to -0.077 in the remaining 675 societies. A global mechanism that reverses when three major ancestry blocs are removed has not traveled.

The gate ledger is therefore unambiguous:

Gate	Result	Meaning
Provenance	Pass	Exact source commits and input hashes match; pinned source worktrees are clean.
Data integrity	Pass	Registered supports, duplicate exclusion, finite matrices, complete outcome levels, and sample threshold satisfied.
Adjusted association	Fail	Family interval includes zero despite the positive point estimate and permutation result.
Coastal separation	Fail	The redundant implementation check includes distance to coast, but its parent gate fails.
Subsistence specificity	Pass	Fishing exceeds hunting and gathering in the registered, composition-dependent comparisons.
Compiler pattern	Fail	Better M2 fit arrives with no fishing-coefficient attenuation.
Transport	Fail	The sign reverses after the three largest language families are excluded.
Ordinal stability	Pass	All four threshold models converge and retain the positive sign.

Four gates passed. Four necessary gates failed. The verdict is rejected at the descriptive claim tier.

This does not mean marine ecology has no political consequences. It means dependence on aquatic calories is too crude and too ancestry-sensitive to serve as a general cause of jurisdictional hierarchy.

V. Fish is a resource topology before it is a nutrient

The failure tells us what the exposure forgot.

EA003 groups fishing, shellfishing, and the hunting of large aquatic animals into a percentage of subsistence. It does not record whether the resource is a predictable salmon run, a dispersed shellfish bed, an offshore hunt, a river fishery, or a seasonal fallback. It does not record storage technology, boat range, ownership, labor organization, market access, or the ability of one coalition to monopolize a harvest site.

Those differences may matter more politically than the shared nutrient label.

Smith and Coddling's study of 89 Indigenous societies along the Pacific Coast of North America provides the strongest concrete example. Their outcome is institutionalized inequality, measured through ranking, unequal food access, and slavery. Their models account for historical and spatial dependence. Reliance on clumped aquatic resources and ownership of fishing sites predict hierarchy, while population pressure, warfare, and scalar stress show much weaker relationships. Their proposed mechanism is not omega-3. It is economic defensibility: a resource concentrated in a predictable place and time can be controlled, and control can be exchanged for labor and allegiance (Smith and Coddling 2021).

That is a political mechanism because it identifies the carrier and the control surface. The resource is carried by a recurring run. Claims are carried by kinship and ownership. Surplus is carried by storage. Hierarchy is carried by rules governing access and distribution.

The broader D-PLACE literature reaches a compatible but less political result. Ahedo and colleagues show that fishing appears across viable mixed subsistence economies and can support nonagricultural combinations associated with sedentism, density, and technological specialization. They also emphasize ecological diversity and the limits of extrapolating Ethnographic Atlas patterns to market economies (Ahedo et al. 2021).

Put simply, “fish” is not one treatment.

A predictable, storable, monopolizable salmon run and a dispersed low-yield fishery may deliver similar nutrients while producing different bargaining structures. A rich fishery can support autonomy when access is open, hereditary inequality when access is controlled, or trade specialization when transport links a surplus to distant demand. Resource properties alter a set of political possibilities. Social rules select among them.

This is the second proof gate. If the same nutrient exposure can support incompatible institutional outcomes, the nutrient cannot be the sufficient cause. The mechanism has to pass through the organization of access, labor, storage, exchange, and enforcement.

VI. Fish and ships are different variables

The original theory also conflated marine nutrition with maritime connectivity.

A coast can provide food. It can also transform distance.

Once a society can travel reliably by water, a body of water becomes a transportation surface, an information network, a military corridor, and a commercial infrastructure. Shipbuilding creates demands for specialized labor and material supply. Voyages create problems of risk sharing, credit, timing, measurement, and agency. Ports create taxable chokepoints. Repeated routes carry news, techniques, pathogens, legal forms, and people. None of these effects requires fish consumption to be the operative variable.

Historical evidence for this network channel is far stronger than evidence for a fish-to-cognition channel. Pascali exploits the transition from sail to steam. Because sail routes depended on wind while steam routes did not, the technology changed effective shipping distances asymmetrically across country pairs. Steam reshaped trade, but its development effects were heterogeneous. Countries with more inclusive institutions benefited; many others did not (Pascali 2017).

Acemoglu, Johnson, and Robinson make a related argument about Atlantic trade after 1500. Access to Atlantic commerce mattered where merchant groups could translate economic gains into institutional changes, especially where preexisting arrangements already constrained monarchy. Ocean access did not act alone. It interacted with political organization (Acemoglu, Johnson, and Robinson 2005).

These studies reject geographical innocence and geographical destiny at the same time. Shipping technology can create a real shock to reachable partners and trade costs. The resulting wealth and power do not distribute themselves. Institutions determine who can enter, contract, tax, expropriate, invest, and preserve gains.

The British Neolithic provides a useful negative control against the nutritional version of the story. Stable-isotope evidence shows a rapid shift from marine to terrestrial diets at the adoption of agriculture, including among coastal populations (Richards, Schulting, and Hedges 2003). Political scale and social complexity did not require a monotonic rise in marine consumption. Farming supported other forms of settlement, storage, population density, and hierarchy while marine dietary reliance fell.

The historical object is therefore not marine exposure in the singular. At minimum, it separates into two variables:

$$N_{it=\text{marine nutritional exposure}}$$

and

$$C_{it} = \text{maritime network connectivity.}$$

Study I measures a rough version of the first and none of the second. A fishing coefficient cannot tell us what a shipping network did. Treating the two as interchangeable would build the desired conclusion into the exposure.

VII. The ecological compiler

We now have enough pieces to name the surviving theory.

An **ecological compiler** is the coupled process by which a population converts environmental constraints and affordances into persistent collective capability.

The idea has four parts.

First, ecology defines a problem and a feasible set. Ultraviolet radiation, temperature, pathogens, resource seasonality, coastlines, rivers, and winds change what bodies and groups must solve. They do not supply a unique political answer.

Second, populations realize solutions through partially substitutable channels. Genes, development, diet, shelter, clothing, migration, storage, navigation, property rules, markets, and public administration can each absorb a different part of the constraint.

Third, useful capacities require carriers. A navigation technique can die with a sailor or persist through apprenticeship, diagrams, instruments, routines, and schools. A productive fishery can remain open, become household property, or be monopolized by a lineage. A trading advantage can dissipate, be taxed away, or fund durable organizations. The carrier determines what survives turnover in individual people.

Fourth, retention is selective. Institutions preserve some practices and suppress others. They decide whose knowledge is legible, whose contract is enforceable, whose risk is pooled, whose route is defended, and whose surplus is captured. Persistence can be productive or predatory. A compiler can retain domination as effectively as it retains discovery.

The full historical state can be represented as

$$S_t = (G_t, B_t, D_t, K_t, N_t, O_t, I_t, R_t, E_t),$$

where genetic distributions, embodied states, diet, knowledge, networks, organizations, institutions, resources, and ecology change together. The next state is not a function of diet alone:

$$S_{t+1} = F(S_t, U_t, \eta_t),$$

with identifiable shocks (U_t) and unresolved contingency (η_t) .

Marine ecology enters the environmental term. Fish consumption enters diet. Nutritional status affects embodied capacity. Boats enter resources and knowledge. Crews, ports, and routes alter networks and organizations. Merchant law, taxation, naval administration, and property rights enter institutions. Those institutions feed back into migration, diet, technology, and survival.

This formulation is not useful because it contains everything. It is useful only when it forces a separation. For any persistence claim, we must ask:

1. What capacity is supposed to persist?
2. What material or social carrier preserves it?
3. What event would break one carrier while leaving another intact?
4. What observation would distinguish nutrition, connectivity, demography, and institutional retention?

That is the carrier problem from *A New Political Science* turned into an empirical discipline.

Consider a port city after a migration shock. If techniques persist despite population replacement, the carrier may be records, infrastructure, or organizations rather than ancestry. If capability disappears when a merchant network is expelled despite unchanged geography, the network was doing causal work. If a shipping innovation expands trade only under enforceable contracts, institutions are part of the treatment. If a nutritional biomarker improves without changes in learning or organization, the biological constraint was not the binding political bottleneck.

The compiler language also keeps biology in its proper place. Bodies matter because all action is embodied. Severe deficiency can shrink the reachable set of attention, learning, health, and effort. But collective capability becomes durable when it is externalized. A chart is remembered navigation. A ledger is remembered exchange. A law is remembered settlement. A school is remembered instruction. An archive is remembered correction.

A civilization is not an organism with a dopamine level. It is a contested persistence architecture built by organisms who die.

VIII. What is now rejected

The research permits several conclusions stronger than agnosticism.

First, EPA and DHA are not precursors of dopamine. Any argument that uses precursor chemistry to move from fish to motivation is wrong at the biochemical step.

Second, current randomized human evidence does not support a large, general cognitive enhancement from omega-3 intake above adequacy. Selective or deficiency-dependent effects remain possible, but they cannot carry a claim about population-level state formation.

Third, fishing dependence does not robustly predict jurisdictional hierarchy across the D-PLACE sample. The adjusted coefficient is positive and passes a within-macroregion permutation test, but the language-family and spatial intervals include zero. The sign reverses when three large language families are removed. The registered claim therefore fails.

Fourth, the European framing is not supported. The adjusted coefficient remains positive outside Europe, and high marine dependence has existed across populations with radically different political forms. Nothing in the reviewed biological or cross-cultural evidence establishes inherited European cognitive superiority.

Fifth, coastline, fish consumption, and maritime connectivity are not interchangeable. A design that measures one cannot claim the effect of another.

These are not rhetorical cautions. They are rejected arrows in the original causal chain.

IX. What survives

Three propositions remain worth testing.

The first is **constraint relaxation**. Marine foods can matter greatly when they relieve a binding nutritional constraint. That effect should be largest under deficiency and weaker around adequacy. It should be measured through biomarkers, development, and task performance rather than inferred from ancestry or historical achievement.

The second is **resource topology**. Aquatic resources differ in clumping, predictability, storability, defensibility, labor requirements, and ownership. Those properties can change bargaining power and political hierarchy. The Smith and Coddling result shows what a credible mechanism looks like: a specified resource distribution, a specified control relation, and a bounded population comparison.

The third is **network conversion**. Maritime technology can change effective distance and the set of reachable partners. Historical consequences depend on institutions that distribute access, enforce exchange, mobilize revenue, and preserve knowledge. Pascali's steamship design demonstrates that connectivity and institutional conversion interact.

Study I does not validate this weaker theory; it clarifies what the next test must measure. Settlement, community size, and population substantially improve model fit, yet they do not attenuate fishing's coefficient on the 467-society complete-case

sample. The proposed social pathway therefore fails its registered pattern gate. If aquatic subsistence matters through organization, the carriers have to be observed directly rather than inferred from a cross-sectional coefficient.

The next decisive study should therefore separate fish from ships. Historical port and voyage records can produce measures of route centrality, reachable ports, voyage frequency, and shipping-time shocks. Fish cargo or local dietary data can measure nutrition separately. Outcomes can include urbanization, book production, schooling, administrative hierarchy, tax capacity, and organizational survival. The identifying tests should use changes in effective distance that do not mechanically change local fish availability, and resource shocks that change marine diets without changing network position.

The biological study should be equally strict. Ancient DNA and isotope evidence can test whether low UV and marine diet interact with pigmentation, vitamin-D, and lipid-metabolism trajectories. Population replacement and admixture must be modeled as rival explanations. The outcome should never be a civilizational rank.

The ecological compiler survives only if its carriers can be observed. Otherwise it becomes a vocabulary for redescribing any outcome after the fact.

Conclusion: the sea as possibility, not destiny

The original theory was searching for a way to connect sunlight, diet, brains, ships, institutions, and historical memory. It found real objects but arranged them in the wrong order.

Low ultraviolet environments created adaptive pressures. Marine foods sometimes relieved them. Omega-3 status can affect neural biology, but EPA and DHA do not become dopamine, and more fish does not become more civilization. The global cross-cultural test rejects fishing dependence as a robust general predictor of political hierarchy. The historical evidence points elsewhere: toward the distribution of resources, the geometry of transport, and the institutions that convert temporary opportunities into durable capacities.

The sea matters because it changes more than nutrition. It changes what can be reached, stored, defended, monopolized, exchanged, learned, and remembered. Those possibilities are politically indeterminate until people organize them.

That is the ecological compiler. Ecology supplies a constraint and an affordance. Bodies make some responses possible. Technologies alter the reachable set. Networks circulate practices. Organizations stabilize routines. Institutions select what persists. History begins to accumulate when capability survives its carrier.

The conclusion is not that fish built Europe. The conclusion is that environments become history through the systems that remember how to use them.

Methods and reproducibility note

The preregistration, amendments, code, tests, reduced public results, and figure are stored under `experiments/ecological_compiler/`. Raw D-PLACE repositories and resampling draws remain under `gitignored/artifacts/ecological_compiler/`. The current Ethnographic Atlas source is pinned to commit `5aa46eea62815daa283ac67cc757065a1b3be16e`; D-PLACE support data are pinned to `9bfed2c8c206be00f55f71516f262bbca2234e5a`. The final result JSON has SHA-256 digest `518d34346930db48a8723615ca75de6db26902828789c73a6f3fb486770f040d`.

The registered run used Python 3.12 with NumPy 2.5.1, SciPy 1.18.0, and Matplotlib 3.11.0, seed 20260826, 300 successful family blocks, 300 successful spatial blocks, and 500 successful within-macroregion permutations. Numerical amendments were logged before the first complete result: analytic gradients were verified against central finite differences, solver ceilings were raised after convergence-limit diagnostics, the D-PLACE Europe suffix mapping was corrected, and all eight registered gates were made noncompensatory in code. A post-result implementation audit then found that the initial code had substituted 50 fine Ethnographic Atlas regions for the six registered D-PLACE macroregions. The final result corrects that representation, isolates missing family labels into singleton clusters, and makes provenance and duplicate checks fail closed. The amendment log preserves what was visible at every repair.

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